SMARTBRIDGE
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SKILL WALLET
A SmartBridge Product

Coding & Solution

Date	11 July 2026
Team ID	
Project Name	Rising waters - Flood Prediction using Machine Learning
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/uttam2309/Rising-Waters-
Programming Language(s)	Python,HTML,CSS,JavaScript
Framework(s) Used	Flask,Scikit-Learn,Pandas,NumPy
Key Features Implemented	• Flood prediction using Machine Learning • User-friendly web interface • Weather & rainfall data input • Real-time flood risk prediction • Trained ML model integration • Responsive frontend
Pending / Incomplete Features	• Live weather API integration • River water level sensor integration • SMS/Email flood alerts • Interactive GIS map visualization
Setup / Run Instructions	• Clone the repository. • Install dependencies using pip install -r requirements.txt. • Run app.py. • Open http://127.0.0.1:5000 in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture with separate frontend, backend, and machine learning components. The application is deployed successfully and designed for future enhancements such as real-time weather APIs, IoT sensor integration, and GIS-based flood monitoring